

The empirical recurrence for OEIS A237923: recovering a conjecture that is not written down

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Abstract

OEIS A237923 records “Empirical recurrence of order 93”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1632$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 93, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 93 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237923 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum plus the lower median of every 2×2 subblock differing from its horizontal and vertical neighbors by exactly one.”. It has offset 1 and begins

1632, 15260, 142488, 1365710, 13009850, 123407764, 1173616996, 11179776830, ...

The entry states:

Empirical recurrence of order 93 (see link above).

As of the “Last modified” line on the live entry (Feb 08 2026 20:44:56, revision 9) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1632$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1632$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 3264$ exact terms of the model returns order 93 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	0	c_{25}	517757385306915	c_{49}	322145141449763379683	c_{73}	−312394965135
c_2	0	c_{26}	3961019889492879	c_{50}	17279391652482266961	c_{74}	−43064362551
c_3	341	c_{27}	10078320453149449	c_{51}	100166241908454936911	c_{75}	192216258795
c_4	3913	c_{28}	4162188939891640	c_{52}	−556078217217935678510	c_{76}	−27418383271
c_5	13523	c_{29}	−36914969960833736	c_{53}	−935416706049692926305	c_{77}	−27708869766
c_6	39886	c_{30}	−105310227687696422	c_{54}	−166456924939482404841	c_{78}	325152921186
c_7	−284677	c_{31}	−136937457100040486	c_{55}	756763819655604730698	c_{79}	67809897123
c_8	−3081154	c_{32}	167166808399387619	c_{56}	2333025224583891413162	c_{80}	−10801827761
c_9	−10463080	c_{33}	866031731860182978	c_{57}	−2013060018261543019009	c_{81}	37197076868
c_{10}	−18755270	c_{34}	1260796485502852392	c_{58}	−3438084483931586106133	c_{82}	35736202879
c_{11}	138778413	c_{35}	259470963946733409	c_{59}	3117439180527712927476	c_{83}	−9727845198
c_{12}	1040870114	c_{36}	−4564566874569190169	c_{60}	−2472353529446645401018	c_{84}	−1679997934
c_{13}	2723239276	c_{37}	−8973297067300673529	c_{61}	2963473660175648380888	c_{85}	14278945789
c_{14}	1682844679	c_{38}	−4068229594948556498	c_{62}	18229708943344828794383	c_{86}	−354211947
c_{15}	−36653218579	c_{39}	11154641973916429330	c_{63}	−7931408715732970115385	c_{87}	−574718755
c_{16}	−172859992683	c_{40}	41196854235032128691	c_{64}	−24052796702508745998016	c_{88}	250058262
c_{17}	−284491054855	c_{41}	32459119042729898385	c_{65}	4684689717842445044440	c_{89}	−210138520
c_{18}	339538438390	c_{42}	−36319575365028456820	c_{66}	5230294765847178418631	c_{90}	−127042106
c_{19}	4920346066098	c_{43}	−67260707204465606203	c_{67}	−5470263297851986791993	c_{91}	103118410
c_{20}	13904306018513	c_{44}	−118629676437682429545	c_{68}	11606891685575214475156	c_{92}	29664849
c_{21}	7596232962001	c_{45}	−21857084672463023904	c_{69}	8604800176831373762829	c_{93}	−2355521
c_{22}	−65425491712873	c_{46}	193986924454251182596	c_{70}	−7418619385527805011528		
c_{23}	−321445313842205	c_{47}	−22462136597586390710	c_{71}	−2469095507996471550104		
c_{24}	−495706608465902	c_{48}	65854319773084939297	c_{72}	967702948094682318687		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 93, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $93 + 4$ terms removed returns order 93 as well, so $\deg q_{\min} = 93 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $93 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 93 that this sequence satisfies — and that family turns out to have exactly one member.

References

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